

Discussion of A Brief Survey of Modern Optimization for Statisticians

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Optimization methods have always played a key role in the development of statistical methodology, but have become of critical importance for modern methods to analyze large size high-dimensional data. Lange, Choi and Zhua (henceforth LCZ) are to be commended for providing a comprehensive overview of optimization methods widely used in statistics. The paper discusses classical unconstrained optimization algorithms (steepest descent and variants), the majorization-maximization framework that has proved very useful in devising novel algorithms for a variety of statistical problems, and also provides a flavor of constrained optimization problems arising in high-dimensional statistics (e.g. regularization, matrix completion, etc.). The topics discussed and their accompanying examples focus on important classes of algorithms that have helped statisticians develop and solve complex models.

In this note, we focus on a class of algorithms suitable for high-dimensional constrained optimization problems. The class of interest is that of *proximal* algorithms, that are very generally applicable in constrained non-smooth optimization, but are in particular very well-suited to recent statistical techniques developed for the analysis of high-dimensional data [1, 10, 13]. In our discussion, we make connections to a number of issues discussed in LCZ, including connections to MM and classical gradient descent algorithms and acceleration schemes.

Proximal Algorithms: We start by providing some definitions and then discuss special cases arising in high-dimensional statistics.

Let $f : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ be a closed proper convex function; i.e. its epigraph is a non-empty closed convex set. The *proximal operator* of f is defined as

$$\text{prox}_f(u) = \underset{x}{\operatorname{argmin}} \left(f(x) + \frac{1}{2\lambda} \|x - u\|_2^2 \right), \quad x, u \in \mathbb{R}^n, \quad (1)$$

where $\|\cdot\|_2^2$ denotes the usual Euclidean norm and $\lambda > 0$ is a parameter that controls the degree to which the proximal operator maps points closer to the minimum of f . As shown

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in [5], the function minimized is strongly convex and not everywhere infinite and hence admits a unique minimizer. Note that the definition implies that $\text{prox}_f(u)$ is a point that approximately minimizes f , but remains close to u .

In general, proximal mapping requires solving the minimization problem posited in (1), but there are instances where the proximal operator admits an explicit expression.

1. $f(x) = 0$, then $\text{prox}_f(u) = u$.
2. $f(x) = I_C(x)$ where I_C denotes the indicator function of a convex set C ; then, $\text{prox}_f(u) = P_C(u)$ corresponds to the *projection* P_C on C .
3. $f(x) = \psi\|x\|_1$, where $\|\cdot\|_1$ denotes the ℓ_1 norm, then the i -th coordinate of $\text{prox}_f(u)$ is given by: (i) $\text{prox}_f(u)_i = 0$ if $|u_i| \leq \psi$, (ii) $\text{prox}_f(u)_i = u_i - \psi$ if $u_i > \psi$ and (iii) $\text{prox}_f(u)_i = u_i + \psi$ if $u_i < -\psi$. It can be seen that the proximal mapping corresponds to a soft-thresholding operation.

A proximal minimization algorithm is defined at the k -th iteration step by

$$x^{k+1} := \text{prox}_f(x^k).$$

As presented in [2], if f has a minimum, then x^k will converge to the set of minimizers of f and $f(x^k)$ to its optimal value. Similarly to steepest descent algorithms, convergence is guaranteed for values of the parameter λ that satisfy $\lambda^k > 0$ and $\sum_{k=1}^{\infty} \lambda^k = \infty$.

This basic algorithm has not found many applications, but is instructive from a theoretical point of view. One important application though, is on solving the quadratic function

$$f(x) = \frac{1}{2}x'Ax - b'x, \quad A \in \mathbb{R}^{n \times n}, \quad x, b \in \mathbb{R}^n$$

with $A \succeq 0$ (positive semidefinite). The proximal mapping can be written down explicitly as

$$\text{prox}_f(u) = \left(A + \frac{1}{\lambda}I\right)^{-1} \left(b + \frac{1}{\lambda}u\right),$$

which gives rise to the update [8]

$$x^{k+1} = \left(A + \frac{1}{\lambda}I\right)^{-1} \left(b + \frac{1}{\lambda}x^k\right).$$

The power of proximal mappings is revealed in the following setting.

Proximal Gradient Method: Consider the following optimization problem

$$\min_x f(x) + g(x), \tag{2}$$

where $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $g : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ are closed proper convex functions and f is *differentiable*. The function g can be used to encode constraints on x as discussed in the “Augmented Lagrangians” section of LCZ.

Then, the proximal gradient method is given by

$$x^{k+1} = \text{prox}_g(x^k - \lambda^k \nabla f(x^k)), \quad (3)$$

where $\lambda^k > 0$ denotes the step size. If $\nabla f(x)$ is Lipschitz continuous with constant M , then this algorithm converges in $\mathcal{O}(1/k)$ steps, for fixed step size $\lambda^k \equiv \lambda \in (0, 1/M)$. If M is unknown, then various line search strategies (akin to those used in gradient descent methods) can be employed.

Note that using the special form of the proximal operator, it can be seen that when $g = 0$ this algorithm reduces to the standard gradient descent one, when $g = \psi\|x\|_1$ (a lasso penalty) it leads to soft-thresholding and for $g = I_C$ it reduces to the projected gradient method [4].

In [3], the proximal gradient algorithm is interpreted as a majorization-minimization algorithm. Consider an upper bound on $f(x)$ given by

$$Q_\lambda(x, y) = f(y) + \nabla f(y)'(x - y) + \frac{1}{2\lambda}\|x - y\|_2^2, \quad \lambda > 0.$$

For fixed λ , $Q_\lambda(x, y)$ is convex and satisfies $Q_\lambda(x, x) = f(x)$ and is an upper bound on f when ∇f is Lipschitz continuous with constant M and $\lambda \in (0, 1/M]$. Then, the algorithm that uses updates of the form

$$x^{k+1} = \text{argmin}_x Q_\lambda(x, x^k)$$

is a majorization-minimization one. Analogously, for the function $f(x) + g(x)$ we can use $\tilde{Q}_\lambda(x, y) = Q_\lambda(x, y) + g(x)$ and some algebra shows that the majorization-minimization based update

$$x^{k+1} = \text{argmin}_x \tilde{Q}_\lambda(x, x^k)$$

is equivalent to (3).

Accelerated Proximal Gradient Method: Nesterov [12] introduced a sequence of updates that *accelerate* the convergence rate from linear ($\mathcal{O}(1/k)$) to quadratic ($\mathcal{O}(1/k^2)$) in convex programming problems. The sequence extrapolates between previous updates of the algorithm. Specifically, for the proximal gradient method it takes the form

$$y^{k+1} = x^k + \omega^k(x^k - x^{k-1}), \quad x^{k+1} = \text{prox}_g(y^{k+1} - \lambda^k \nabla f(y^{k+1})), \quad (4)$$

where $\omega^k \in [0, 1)$ is an extrapolation parameter and λ^k the usual step size. A simple choice for the extrapolation parameter is $\omega^k = k/(k+3)$.

Then, it can be established that for ∇f Lipschitz continuous with constant M , the acceleration scheme guarantees convergence in $\mathcal{O}(1/k^2)$ steps with a fixed step size $\lambda^k \equiv \lambda \in (0, 1/M]$. As before, if M is unknown the step sizes can be determined through a line search.

Stochastic proximal gradient algorithms: An area where we believe more research is needed to specifically suit the needs of statistics is stochastic optimization. This corresponds for example to (2), where the function f is given as an intractable integral of the

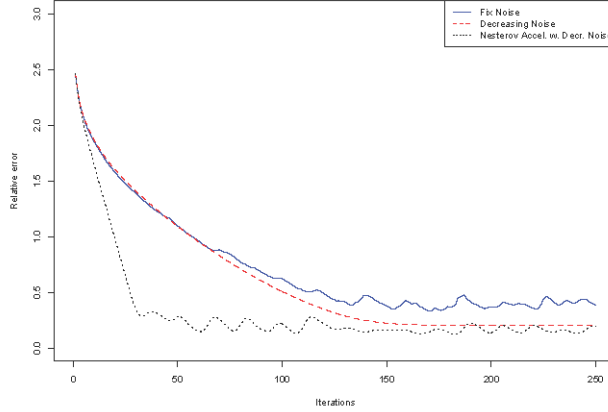


Figure 1: Relative errors for stochastic proximal gradient algorithm for a random effects logistic model with $p = 100$, $T = 10$, $k = 200$.

form $f(x) = \int F(x, \zeta) \Pi(d\zeta)$. This is a well-known problem in stochastic programming and online learning and has generated various stochastic extensions of the algorithms presented above ([11, 14, 6, 9]). But in statistics this problem takes a more challenging form. Latent variables abound in statistics and leads to log-likelihood functions and their derivatives that are intractable: $f(x) = \log \int F(x, \zeta) \Pi(d\zeta)$, $\nabla f(x) = \int \nabla_x \{\log F(x, \zeta)\} \check{\Pi}_x(d\zeta)$, where $\check{\Pi}_x(d\zeta) \propto F(x, \zeta) \Pi(d\zeta)$. The important point here is that the distribution $\check{\Pi}_x$ is typically very difficult to simulate, and unlike most examples in online learning stochastic optimization, $\check{\Pi}_x$ depends on x . Nevertheless, one can easily adapt the proximal gradient algorithm and its accelerated version presented above, by replacing $\nabla f(x)$ by a Monte Carlo approximation $\widehat{\nabla} f(x)$ obtained by simulation from $\check{\Pi}_x$ (possibly using Markov Chain Monte Carlo methods). The approximation can be obtained with a fixed number of Monte Carlo steps (called fixed noise below), or with an increasing number of Monte Carlo steps (decreasing noise).

For illustration purposes, consider the following random effects logistic regression example. We have N statistical units with repeated binary responses $\{y_{it}, 1 \leq t \leq T_i\}$, $y_{it} \in \{0, 1\}$. For the covariate matrix $Z \in \mathbb{R}^{n \times p}$, with i -th row denoted by z_i , we assume that

$$y_{it}|u_i \sim \text{Ber} \left(\frac{e^{z_i \beta + u_i}}{1 + e^{z_i \beta + u_i}} \right), \quad 1 \leq i \leq n, \quad 1 \leq t \leq T_i,$$

where $u_i \stackrel{i.i.d.}{\sim} N(0, \sigma^2)$. The log-likelihood of β is intractable and requires integrating out the random effect u . We estimate β by an ℓ^1 -penalized likelihood approach; thus, $g(\beta) = \psi \|\beta\|_1$.

We implemented the proximal gradient algorithm, and its accelerated version by replacing in (3) and (4) $\nabla f(x^k)$ by $\widehat{\nabla} f(x^k)$ obtained using a Markov Chain Monte Carlo scheme. Figure 1 depicts the relative error $\|\beta_k - \beta_\star\| / \|\beta_\star\|$, as a function of the iterations k for both algorithms, where $\beta_\star$ denotes the true value of the parameter vector.

This simulation example suggests that stochastic versions of the proximal gradient algorithm and its accelerated versions can be designed to deal with high-dimensional statistical models with intractable log-likelihood functions. However, more work is needed to gain a clear and deep understanding of the conditions needed so that these extensions work and exhibit convergence properties analogous to their deterministic counterparts.

Alternating Direction Method of Multipliers: In the proximal gradient algorithm the function f was assumed to be smooth. However, the method can be adapted to handle the following variant of the minimization problem:

$$\min_x f(x) + g(x),$$

where $f, g : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ are closed proper convex functions and both f, g can be *non-differentiable*. Then, the alternating direction method of multipliers employs the following updates:

$$x^{k+1} = \text{prox}_f(z^k - u^k), \quad z^{k+1} = \text{prox}_g(x^{k+1} + u^k), \quad u^{k+1} = u^k + x^{k+1} - z^{k+1}.$$

It can be seen that this method handles the two functions in the objective function completely separately through their proximal operators. It is most useful when the proximal operators of f and g can be easily computed, but that of the objective function $f + g$ is not. The convergence theory for this method is discussed in detail in [7].

References

- [1] Bach, F., Jenatton, R., Mairal, J. and Obozinski, G. (2011), Optimization with sparsity-inducing penalties, *Foundations and Trends in Machine Learning*, 4, 1106
- [2] Bauschke, H. and Combettes, P. (2011), *Convex Analysis and Monotone Operator Theory in Hilbert Spaces*, Springer
- [3] Beck, A. and Teboulle, M. (2012), Smoothing and first order methods: a unified framework, *SIAM Journal on Optimization*, 22, 557-580
- [4] Bertsekas, D. (1999), *Nonlinear Programming*, Athena Scientific
- [5] Boyd, S. and Vandenberghe, L. (2004), *Convex Optimization*, Cambridge University Press
- [6] Duchi, J. C., Agarwal, A., Johansson, M. and Jordan, M. (2012), Ergodic mirror descent, *SIAM Journal of Optimization*, 22, 1549-1578
- [7] Boyd, S., Parikh, N., Chu, E., Peleato, B. and Eckstein, J. (2011), Distributed optimization and statistical learning via the Alternating Direction Method of Multipliers, *Foundations and Trends in Machine Learning*, 3, 1122

- [8] Golub, G. and Wilkinson, J. (1996), Note on the iterative refinement of least squares solution, *Numerische Mathematik*, 9, 139-148
- [9] Lan, G. (2012), An optimal method for stochastic composite optimization, *Mathematical Programming, Series A*, 133:365-397.
- [10] Lee, J., Recht, B., Salakhutdinov, R., Srebro, N. and Tropp, J. (2010), Practical large scale optimization for max-norm regularization, *Advances in Neural Information Processing Systems*, 23, 1297-1305
- [11] Nemirovski, A., Juditsky, A. Lan, G., and Shapiro, A. (2009), Robust Stochastic approximation approach to stochastic programming, *SIAM Journal of Optimization*, 19, 1574-1609
- [12] Nesterov, Y. (1983), A method of solving a convex programming problem with convergence rate $O(1/k^2)$, *Soviet Mathematics Doklady*, 27, 372-376
- [13] Ravikumar, P., Agarwal, A. and Wainwright, M. (2010), Message-passing for graph-structured linear programs: Proximal methods and rounding schemes, *Journal of Machine Learning Research*, 11, 1043-1080
- [14] Xiao, L. (2010), Dual averaging methods for regularized stochastic learning and online optimization, *Journal of Machine Learning Research*, 11, 2543-2596.